ref input, SimPoint estimates — small-dm

L1D 64 sets x 1, L2 512 sets x 1 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

36 of 36 runs completed, 0 running, 0 pending; **0 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	20.65%	47.95%	56.30%	20.67%	48.00%	56.39%
parest	1/1	0.16%	0.43%	0.62%	0.64%	1.19%	7.86%
deepsjeng	1/1	2.92%	9.17%	16.25%	8.11%	23.58%	49.72%
leela	1/1	0.38%	1.36%	1.80%	6.07%	15.90%	45.53%
namd	1/1	3.27%	8.21%	17.87%	3.27%	8.22%	17.90%
lbm	1/1	10.48%	32.28%	44.97%	10.48%	32.28%	44.97%
nab	1/1	5.20%	15.37%	34.02%	6.75%	18.74%	49.89%
xz_cld	1/1	3.38%	12.11%	22.88%	5.08%	17.35%	36.71%
xz_cpu2006docs	1/1	1.47%	5.78%	23.30%	2.03%	7.38%	37.90%
xz_combined	1/1	2.85%	10.66%	26.32%	4.17%	14.65%	42.32%
imagick	1/1	0.23%	0.12%	3.44%	5.33%	12.37%	46.11%
blender	1/1	0.38%	0.88%	3.23%	3.72%	6.96%	42.51%
povray	1/1	0.27%	0.46%	0.95%	14.26%	22.18%	55.24%
omnetpp	1/1	0.00%	0.00%	0.00%	12.75%	24.67%	48.13%
xalancbmk	1/1	0.13%	0.26%	0.80%	4.92%	8.32%	39.84%
x264_pass1	1/1	7.07%	15.02%	38.53%	10.18%	21.51%	55.90%
x264_pass2	1/1	6.25%	13.38%	40.16%	8.63%	18.23%	56.21%
x264_seek	1/1	6.41%	13.66%	41.07%	8.82%	18.56%	57.26%
gcc_pp_O3	1/1	1.23%	2.73%	4.48%	16.24%	33.66%	64.23%
gcc_pp_O2	1/1	1.25%	2.75%	4.47%	16.60%	34.40%	63.73%
gcc_smaller	1/1	1.09%	2.19%	4.61%	13.76%	25.29%	64.09%
gcc_ref32_O5	1/1	0.91%	1.84%	3.40%	15.53%	30.63%	59.73%
gcc_ref32_O3	1/1	0.68%	1.85%	1.06%	11.25%	27.66%	19.94%
mcf	1/1	1.40%	2.06%	7.74%	2.33%	3.60%	12.43%
perlbench_checkspam	1/1	7.98%	9.83%	29.42%	22.33%	36.24%	69.19%
perlbench_diffmail	1/1	1.96%	4.19%	4.64%	17.81%	31.25%	54.38%
perlbench_splitmail	1/1	4.16%	4.19%	15.82%	23.27%	35.82%	68.04%
bwaves_1	1/1	2.51%	6.24%	18.43%	2.51%	6.24%	18.43%
bwaves_2	1/1	2.59%	6.43%	18.99%	2.59%	6.43%	18.99%
bwaves_3	1/1	2.51%	6.23%	18.45%	2.51%	6.23%	18.45%
bwaves_4	1/1	2.56%	6.32%	19.44%	2.56%	6.32%	19.44%
exchange2	1/1	5.23%	10.55%	12.68%	6.08%	12.17%	14.84%
fotonik3d	1/1	3.36%	9.32%	9.25%	3.38%	9.37%	9.40%
roms	1/1	4.37%	13.83%	8.97%	4.44%	13.96%	9.48%
wrf	1/1	0.91%	3.38%	2.59%	1.52%	4.76%	7.57%
cam4	1/1	3.14%	11.02%	11.03%	4.99%	15.77%	21.23%
pooled		3.78%	9.06%	17.53%	7.49%	16.44%	39.26%
mean		3.31%	8.11%	15.78%	8.49%	18.33%	39.00%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	16.94%	6.47%	34.17%	5.01%	16.06%	55.00%	19.96%	15.81%
parest	done	15.19%	10.39%	16.13%	5.80%	6.57%	51.66%	4.51%	9.35%
deepsjeng	done	13.61%	6.44%	17.15%	9.67%	7.29%	14.36%	6.53%	23.10%
leela	done	7.98%	7.04%	8.76%	0.28%	5.79%	17.91%	0.14%	0.36%
namd	done	14.97%	6.99%	18.59%	2.52%	3.62%	32.32%	9.33%	14.52%
lbm	done	22.89%	1.68%	12.45%	5.04%	83.18%	78.54%	3.75%	1.81%
nab	done	13.74%	3.93%	18.49%	2.85%	1.41%	31.34%	1.48%	3.09%
xz_cld	done	15.16%	3.62%	20.80%	2.69%	6.66%	34.07%	3.06%	1.80%
xz_cpu2006docs	done	11.35%	4.71%	12.12%	3.05%	9.83%	41.33%	5.77%	7.82%
xz_combined	done	14.39%	5.20%	17.33%	3.88%	9.21%	28.18%	2.29%	2.00%
imagick	done	2.76%	6.34%	3.43%	1.40%	0.40%	68.78%	40.68%	0.61%
blender	done	10.33%	7.50%	10.97%	3.36%	6.54%	31.65%	4.85%	21.64%
povray	done	14.40%	9.21%	17.67%	22.20%	6.19%	37.67%	32.76%	47.94%
omnetpp	done	19.33%	12.85%	25.22%	2.13%	7.83%	37.69%	0.05%	4.21%
xalancbmk	done	18.92%	13.80%	21.61%	17.84%	5.88%	68.27%	8.55%	24.98%
x264_pass1	done	4.66%	0.69%	5.80%	1.38%	5.07%	24.17%	3.30%	13.63%
x264_pass2	done	4.16%	0.78%	5.00%	1.11%	4.67%	28.76%	4.93%	17.35%
x264_seek	done	4.25%	0.92%	5.09%	1.37%	4.76%	23.46%	3.09%	13.24%
gcc_pp_O3	done	13.49%	7.07%	17.49%	8.26%	6.21%	38.93%	3.57%	15.30%
gcc_pp_O2	done	13.50%	6.93%	17.51%	8.63%	6.45%	37.38%	5.64%	13.67%
gcc_smaller	done	14.57%	6.22%	18.06%	7.17%	6.18%	45.12%	7.65%	10.35%
gcc_ref32_O5	done	13.63%	7.02%	17.52%	8.71%	6.31%	44.76%	6.86%	10.91%
gcc_ref32_O3	done	13.94%	9.32%	18.06%	10.74%	11.04%	62.89%	12.68%	19.94%
mcf	done	24.46%	11.70%	30.85%	9.98%	5.87%	69.97%	9.12%	20.77%
perlbench_checkspam	done	6.93%	2.21%	10.27%	3.55%	2.91%	19.65%	1.52%	2.59%
perlbench_diffmail	done	8.95%	2.24%	11.37%	12.13%	4.88%	29.10%	2.49%	7.67%
perlbench_splitmail	done	13.58%	1.38%	17.71%	8.34%	7.81%	14.31%	1.51%	2.69%
bwaves_1	done	44.41%	7.12%	48.35%	1.07%	38.47%	90.83%	8.93%	1.22%
bwaves_2	done	44.03%	7.03%	48.06%	1.14%	38.10%	91.21%	8.45%	1.31%
bwaves_3	done	45.97%	7.16%	49.90%	1.06%	41.59%	41.97%	19.35%	1.47%
bwaves_4	done	44.01%	7.08%	47.99%	1.21%	37.57%	44.00%	18.07%	0.98%
exchange2	done	3.11%	1.88%	3.96%	2.01%	2.31%	0.55%	0.15%	0.52%
fotonik3d	done	22.00%	9.79%	28.97%	0.50%	5.16%	49.28%	13.24%	3.95%
roms	done	30.33%	6.83%	37.70%	3.25%	19.59%	40.55%	15.25%	5.82%
wrf	done	26.09%	13.66%	27.65%	8.20%	22.54%	29.03%	13.89%	17.61%
cam4	done	15.12%	5.77%	16.68%	4.62%	15.55%	47.00%	9.17%	18.53%
pooled		15.65%	5.43%	19.22%	4.14%	9.83%	45.16%	14.40%	13.25%
mean		17.31%	6.36%	20.52%	5.34%	13.04%	41.71%	8.68%	10.52%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save

column	formula	aciklama
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.